Cloud Computing TASK – 1

1. **Log in to your AWS account, open the VPC dashboard, and create a new VPC named InstaVibeVPC with a custom IPv4 CIDR block (e.g., 10.0.0.0/16).**

   Ans :

      - Sign in to the AWS Management Console.
      - In the search bar, type VPC and open the VPC Dashboard.
      - In the left navigation pane, select Your VPCs.
      - Click Create VPC.
      - Under Resources to create, select VPC only.

   - Enter the following details:
   - Name tag: InstaVibeVPC
   - IPv4 CIDR block: 10.0.0.0/16
   - IPv6 CIDR block: No IPv6 CIDR block (unless your assignment requires IPv6)
   - Tenancy: Default
   - Click Create VPC.
      - Wait for the confirmation message indicating the VPC has been created successfully.

2. **Inside your InstaVibeVPC, create two subnets: one public subnet (e.g., 10.0.1.0/24) for hosting a web server and one private subnet (e.g., 10.0.2.0/24) for backend services.

Hint: Name your subnets as 'PublicSubnet-Zomato' and 'PrivateSubnet-Zomato' for clarity.**

   Ans :

   Step 1: Open the Subnets Page

      - In the AWS Management Console, go to VPC Dashboard.

- In the left navigation pane, click Subnets.
- Click Create subnet.

Step 2: Select Your VPC

- VPC ID: Select InstaVibeVPC.

Step 3: Create the Public Subnet

Setting	Value
Subnet name	PublicSubnet-Zomato
Availability Zone	Choose any AZ (e.g., us-east-1a)
IPv4 VPC CIDR block	10.0.0.0/16
IPv4 subnet CIDR block	10.0.1.0/24

Step 4: Create the Private Subnet

Setting	Value
Subnet name	PrivateSubnet-Zomato
Availability Zone	Choose the same or another AZ (e.g., us-east-1b)
IPv4 VPC CIDR block	10.0.0.0/16
IPv4 subnet CIDR block	10.0.2.0/24

- Click Create subnet.

3. **Configure an Internet Gateway and attach it to your InstaVibeVPC so that only the public subnet can access the internet, while the private subnet cannot.**

Ans :

Step 1: Create an Internet Gateway

- o Open the VPC Dashboard.
- o In the left navigation pane, select Internet Gateways.
- o Click Create internet gateway.

o Enter:

- o Name tag: InstaVibe-IGW
- o Click Create internet gateway.

Step 2: Attach the Internet Gateway to the VPC

- o Select the newly created InstaVibe-IGW.
- o Click Actions → Attach to VPC.
- o Select InstaVibeVPC.
- o Click Attach internet gateway.

Step 3: Create a Route Table for the Public Subnet

- o In the VPC Dashboard, select Route Tables.
- o Click Create route table.

o Enter:

- o Name: PublicRouteTable
- o VPC: InstaVibeVPC
- o Click Create route table.

Step 4: Add a Route to the Internet Gateway

- o Select PublicRouteTable.
- o Open the Routes tab.
- o Click Edit routes → Add route.
- o Configure:
- o Destination: 0.0.0.0/0
- o Target: Select Internet Gateway → InstaVibe-IGW
- o Click Save changes.

Step 5: Associate the Public Subnet

- With PublicRouteTable selected, open the Subnet associations tab.
- Click Edit subnet associations.
- Select PublicSubnet-Zomato.
- Click Save associations.

4. **Set up route tables so that your public subnet routes traffic to the Internet Gateway, but your private subnet does not. Verify this by checking the route table associations.**

   Ans :

   Step 1: Create a Public Route Table

   - Open the AWS Management Console and go to VPC Dashboard.
   - In the left navigation pane, click Route Tables.
   - Click Create route table.
   - Enter:
   - Name: PublicRouteTable
   - VPC: InstaVibeVPC
   - Click Create route table.

   Step 2: Add an Internet Route

   - Select PublicRouteTable.
   - Go to the Routes tab.
   - Click Edit routes → Add route.
   - Configure the route:
   - Destination: 0.0.0.0/0
   - Target: Internet Gateway → InstaVibe-IGW
   - Click Save changes.

   Step 3: Associate the Public Subnet

   - Select PublicRouteTable.
   - Open the Subnet associations tab.
   - Click Edit subnet associations.

- Select PublicSubnet-Zomato.
- Click Save associations.

Step 4: Configure the Private Subnet

- You have two options:
- Option 1 (Recommended): Leave PrivateSubnet-Zomato associated with the Main Route Table, which contains only the local route.
- Option 2: Create a separate route table named PrivateRouteTable, associate it with PrivateSubnet-Zomato, and do not add a 0.0.0.0/0 route to the Internet Gateway.

5. **Draw a simple diagram (hand-drawn or using any online tool) showing your VPC, public subnet, private subnet, and Internet Gateway, and briefly explain in 3-4 lines how this setup is similar to how apps like Zomato separate their public user interface from their private databases.**

Ans :

```
  Internet
      |
+-----------------+
| Internet Gateway|
+-----------------+
      |
+-----------------+
|  InstaVibeVPC  |
|  10.0.0.0/16  |
```

```
          +----------------+
            |         |
   +---------------+   +---------------+
   | Public Subnet  |   | Private Subnet |
   |10.0.1.0/24     |   |10.0.2.0/24     |
   |PublicSubnet-   |   |PrivateSubnet-  |
   |Zomato          |   |Zomato          |
   | Web Server     |   | Backend/DB     |
   +---------------+   +---------------+
        |
   Internet Access ✅
```

Private Subnet: No Internet Access ❌

In this setup, the public subnet hosts the web server and is connected to the Internet through the Internet Gateway, allowing users to access the application. The private subnet contains backend services or databases and has no direct internet access, making it more secure. This is similar to apps like Zomato, where users interact with public-facing web or application servers, while sensitive data such as user accounts, restaurant information, and orders are stored securely in private backend systems.